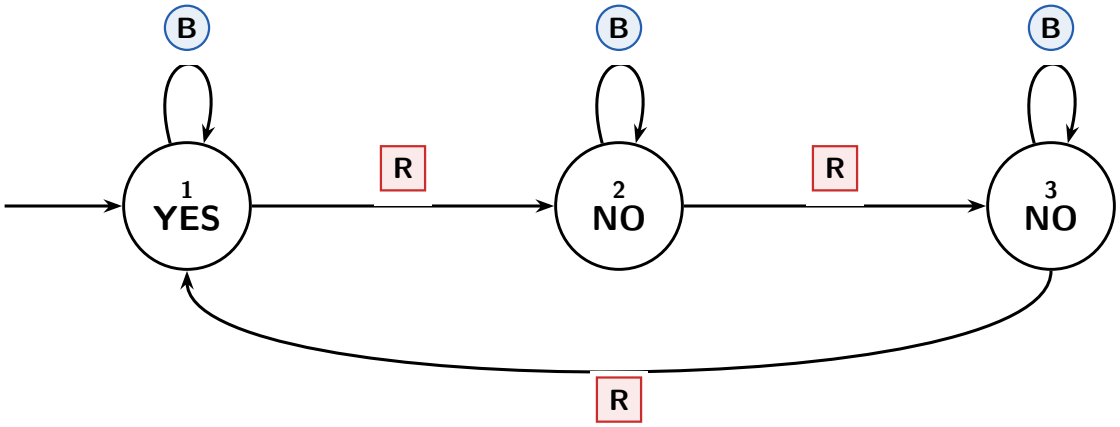
R means red; B means blue. A machine has one start circle and a fixed YES or NO on each circle. Each circle has exactly one R arrow and one B arrow. Read left to right with one marker, hiding used cards. Only the current circle and the card being read may affect the next move.

Problem 1: Find every five-card row this machine sends to YES. Describe exactly which rows it sends to YES, for any length.



Example R B: start 1 $\xrightarrow{R}$ 2 $\xrightarrow{B}$ 2; stop: NO.

Problem 2: A row gets YES exactly when its red cards can be grouped in threes with none left over, including no red cards. For each pair, add the same extra cards in the same order to both rows to give one YES and one NO, or explain why this cannot be done. Adding no cards is allowed.

no cards

R

\+ _____

YES / NO

YES / NO

R

R R

\+ _____

YES / NO

YES / NO

R R

B R R

\+ _____

YES / NO

YES / NO

B B B

R R R

\+ _____

YES / NO

YES / NO
